Replication: First-Generation Elite

The Role of School Social Networks (Cattan, Salvanes, and Tominey, AER 2025): an audit of the IV mediation design and a calibrated substitute for the registry data

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1 The paper in one page

The question is whether going to high school with the children of the elite makes low-income students more likely to enter elite degrees, and the answer is yes, through a channel that runs against a channel through grades. Norwegian registry data follow every student who finished middle school between 2002 and 2010 into high school, university, and the labour market. Exposure to elite peers is the share of a student's high-school cohort whose parents hold an elite degree, standardized. Within-school variation across cohorts identifies the effect, with school, cohort, and program fixed effects. For low-SES students a one standard deviation rise in exposure raises enrollment in an elite degree by 1.1 percentage points (Table 2), about a fifth of the baseline rate.

The mechanism section is an instrumental-variables mediation analysis. Elite peers lower grades (competition, or teacher grading), and grades govern admission, so the total effect hides a negative channel. To separate the channels the authors instrument the high-school grade point average with a lottery: in the third year of high school students are randomly assigned to a written mathematics exam, which is graded blind and raises the GPA of those assigned. Table 4 reports the first stage (GPA on peers and on the lottery), the second stage (enrollment on peers and instrumented GPA), and the decomposition: the direct effect is the second-stage peer coefficient γ_1, the indirect effect is the first-stage peer coefficient times the instrumented GPA effect, $\delta_1\gamma_2$, and the total effect is their sum. For low-SES students the direct effect (0.026) is larger than the total (0.011) because the indirect effect through grades is negative (-0.015).

Why this paper anchors the endogenous-mediator part of Section 10. The mediator is a choice-laden outcome, GPA, and the decomposition needs an exogenous shifter of it. The lottery is an encouragement design for the mediator. The extra assumption, that the effect of GPA on enrollment is the same for the lottery's compliers as for the students whose grades the peers move (Huber 2019), is what the paper's Appendix Figure A13 checks by re-estimating γ_2 across subgroups.

What the replication package holds and does not hold. The registry data cannot be released; the package contains the complete do-files, the auxiliary inputs (crosswalks, CPI series, subject codes, occupation lists), and the README. There is no synthetic dataset. This note audits the code path that produces Table 4, then builds a design calibrated to the published Table 4 and runs `mediate_iv()` on it to reproduce the table's layout, the homogeneity check, and the weak-instrument column.

2 Audit of the replication package

The package is a code map: 22 do-files, no analysis data. `Master_Replication.do` sets the paths to a restricted `replication_data` folder and calls the others in order.

Table 1: The replication package by output and by missing input. Every analysis file is built from Norwegian registries that the package cannot include.

Do-file	Produces	Missing input
Replication_ElitePeers.do, Analysis_Replication.do	Tables 1-3, Figure 1: exposure measure, total effects by SES	elitetemp.dta (analysis file from registries)
Analysis_IV_Replication.do	Table 4 (IV mediation), Table A11 (lottery balance)	tab_vitnemalfag_vgskole (exam registry), elitetemp.dta
FigureA13_Replication.do	Figure A13: gamma2 by subgroup (homogeneity check)	elitetemp.dta
simulate_replication.do	Section VII: reallocation of low-SES pupils across schools and predicted mobility	elitetemp.dta, rankrank_Replication.dta
Analysis_Longrun_Replication.do, RankRank_Replication.do	Tables 5-6: earnings and rank-rank mobility	earnings registries
TeacherData_Replication.do, MeasurementError_Replication.do, Placebo_Replication.do	Appendix: teacher composition, measurement error, placebo cohorts	teacher and census registries

The Table 4 specification is explicit in `Analysis_IV_Replication.do`. The instrument `writtenmaths` is built from the exam registry as an indicator for a written mathematics exam in the third year of a three-year high school. Then, for the low-SES sample:

```
xtivreg2 elitedegree mean_pexecutive_schoolstd $controlsminus spec alternative_educ
    (gpa_tot_sd = writtenmaths) [pweight = meansize2] if parentedu == 1, fe first
```

with `xtset skole_orgnr1` (school fixed effects), cohort dummies and the parental-education, gender, origin, middle-school GPA, and parental occupation controls in `$controlsminus`, high-school program dummies (`spec`, `alternative_educ`), and weights equal to the school size. The same call on the high-SES sample gives columns 3 and 4, and `xtreg ..., fe` without GPA gives the OLS columns. Three observations from the audit:

- The OLS columns of Table 4 are the total-effect regressions without GPA (the Table 2 specification on the IV sample), not OLS with GPA as a regressor. Panel C's "total effect" is that coefficient.
- The IV call carries `[pweight = meansize2]`, which makes `xtivreg2` report heteroskedasticity-robust standard errors; the table note says "clustered at the school level". The cluster option appears in the Table A11 regressions of the same file but not in the Table 4 line.

- Panel C reports no standard errors for the direct, indirect, and total effects. `mediate_iv()` bootstraps the whole system, which supplies them.

`simulate_replication.do` is not a data generator. It is the Section VII exercise: it reallocates low-SES students across schools cohort by cohort to raise their exposure, re-estimates the enrollment and earnings regressions by SES with `parmest`, and predicts outcomes under the new allocation. It reads `elitetemp.dta` on its first line. There is therefore no simulated dataset in the package to run, and the design below is this note's substitute.

3 A design calibrated to Table 4

The design keeps the structure of the paper's identification and the magnitudes of its Table 4. Schools with sizes drawn around the paper's mean (58,304 pupils in 500 schools), nine cohorts, three programs, school-cohort exposure to elite peers standardized across school-cohorts, an unobserved ability that raises both GPA and enrollment (so that OLS on GPA is biased), a binary lottery for the written exam with the published first stage, and school-cohort shocks so that clustering by school matters. The outcome is kept linear so that the coefficients are the parameters of the design.

```
sim_cattan <- function(n_schools = 500, mean_size = 117, delta1 = -0.039, delta2 = 0.094,
                       gamma1 = 0.026, gamma2 = 0.384, kappa = 0.15, sd_sc = 0.25, sd_gpa = 1.3, sd_y = 0.4,
                       seed = 1) {
  set.seed(seed)
  size <- pmax(10, round(rgamma(n_schools, shape = 4, scale = mean_size / 4)))
  n <- sum(size)
  school <- rep(seq_len(n_schools), size)
  cohort <- sample(2002:2010, n, replace = TRUE)
  program <- sample(1:3, n, replace = TRUE)
  school_fe <- rnorm(n_schools, 0, 0.3)[school]
  cohort_fe <- (cohort - 2006) * 0.01
  key <- paste(school, cohort)
  sc_ids <- unique(key)
  peer_raw <- rnorm(length(sc_ids), 0.10, 0.06)[match(key, sc_ids)]
  peer <- (peer_raw - mean(peer_raw)) / sd(peer_raw)
  u_sc_gpa <- rnorm(length(sc_ids), 0, sd_sc)[match(key, sc_ids)]
  u_sc_y <- rnorm(length(sc_ids), 0, sd_sc / 4)[match(key, sc_ids)]
  female <- rbinom(n, 1, 0.5)
  ability <- rnorm(n)
  z <- rbinom(n, 1, 0.5)
  gpa <- school_fe + cohort_fe + delta1 * peer + delta2 * z + 0.1 * female + 0.5 * ability + u_sc_gpa + rnorm(n, 0, sd_gpa)
  y <- 0.15 + 0.5 * school_fe + cohort_fe + gamma1 * peer + gamma2 * gpa + kappa * ability + 0.02 * female + u_sc_y + rnorm(n, 0
  structure(data.frame(y = y, gpa = gpa, peer = peer, z = z, female = female, school = school, cohort = cohort,
                       program = program, size = size[school]),
            truth = list(direct = gamma1, indirect = delta1 * gamma2, total = gamma1 + delta1 * gamma2,
                         first_stage = delta1, mediator_effect = gamma2, lottery = delta2))
}
low <- sim_cattan(seed = 1)
high <- sim_cattan(n_schools = 409, mean_size = 49, delta1 = -0.010, delta2 = 0.043, gamma1 = 0.049, gamma2 = 0.384, seed = 2)
c(low_n = nrow(low), low_schools = length(unique(low$school)), high_n = nrow(high), high_schools = length(unique(high$school)))
#>        low_n  low_schools       high_n high_schools
#>        57820          500        20734          409
```

The high-SES design differs in three published ways. Fewer pupils (19,968 in 409 schools), a smaller first-stage effect of the lottery (0.043 against 0.094, with an F of 5.6 against 76.9), and a larger direct effect (0.049). The published γ_2 for high-SES students is 1.771 with a standard error of 0.714, which a weak instrument produces from the same 0.384 as easily as from anything else; the design keeps 0.384 so that the weak-instrument column can be read against a known truth.

4 Table 4 in the published layout

One call per SES group gives every entry of the table. `mediate_iv()` fits the first stage, the instrumented second stage, and the reduced forms with `fixest`, with the school, cohort, and program fixed effects, the school-size weights, and clustering by school, and bootstraps the system by school for the decomposition's standard errors.

```r
fit_low <- mediate_iv(low, "y", "peer", "gpa", z = "z", x = "female", fe = c("school", "cohort", "program"),
                      cluster = "school", weights = "size", n_boot = n_boot, seed = 1)
fit_high <- mediate_iv(high, "y", "peer", "gpa", z = "z", x = "female", fe = c("school", "cohort", "program"),
                       cluster = "school", weights = "size", n_boot = n_boot, seed = 1)
```

Table 2: Table 4 layout from the calibrated design: Panel A first stage (GPA on peers and the lottery, with the first-stage F), Panel B second stage (enrollment on peers and instrumented GPA; OLS columns are the total-effect regressions without GPA), Panel C the decomposition with school-bootstrap standard errors that the paper does not report. School, cohort, and program fixed effects; weights by school size; standard errors clustered by school.

	Low SES OLS	Low SES IV	High SES OLS	High SES IV
Panel A. First-stage outcome: high school GPA				
Proportion of parents with elite degree (std)		-0.031		0.008
		(0.009)		(0.012)
Student took written math exam (IV)		0.094		0.033
		(0.013)		(0.021)
F-statistic		63.58		2.87
Panel B. Second-stage outcome: enrollment to elite degree				
Proportion of parents with elite degree (std)	0.014	0.025	0.046	0.045
	(0.005)	(0.003)	(0.007)	(0.076)
High school GPA		0.370		0.115
		(0.045)		(4.606)
Panel C. Decomposition				
Direct effect		0.025 (0.003)		0.045 (0.076)
Indirect effect		-0.012 (0.004)		0.001 (0.076)
Total effect		0.014 (0.005)		0.046 (0.007)
Number of pupils	57,820	57,820	20,734	20,734
Number of schools	500	500	409	409

Table 3: Table 4 as published (Cattan, Salvanes, and Tominey 2025), for comparison.

	Low SES OLS	Low SES IV	High SES OLS	High SES IV
Panel A. First-stage outcome: high school GPA				
Proportion of parents with elite degree (std)		-0.039		-0.010
		(0.007)		(0.010)
Student took written math exam (IV)		0.094		0.043
		(0.011)		(0.018)
F-statistic		76.86		5.57
Panel B. Second-stage outcome: enrollment to elite degree				
Proportion of parents with elite degree (std)	0.011	0.026	0.032	0.049
	(0.004)	(0.004)	(0.009)	(0.019)
High school GPA		0.384		1.771
		(0.053)		(0.714)
Panel C. Decomposition				
Direct effect		0.026		0.049
Indirect effect		-0.015		-0.018
Total effect		0.011		0.031
Number of pupils	58,304	58,304	19,968	19,968
Number of schools	500	500	409	409

The low-SES column reproduces the published structure with the published magnitudes, and

the decomposition now carries standard errors. The direct effect is 0.025 (0.003) for a design value of 0.026, the indirect effect -0.012 (0.004) for -0.015, and the total 0.014 for 0.011. The first-stage F is 63.6 against the paper's 76.9; both are the conventional F, which `xtivreg2` prints under `first` and `fixest` reports as `ivf`, and the gap is a matter of how much residual variance the design puts in GPA rather than a difference in definition. The identity total = direct + indirect holds exactly for the reduced form that includes the lottery, which is the `total_z` row of the object; the `total` row without the lottery differs by sampling error only.

The high-SES column shows what a first-stage F of 2.9 does to the decomposition. The instrumented GPA effect is 0.115 with a standard error of 4.606, and the indirect effect's interval covers zero and values several times the design's -0.004. The paper's 1.771 (0.714) is the same phenomenon on the real data. The Anderson-Rubin set from Section 05's `iv_ar_confidence_set()` shows what a weak-instrument-robust interval for γ_2 looks like in this column. The helper takes controls but not fixed effects, so it is run on the high-SES design without the school, cohort, and program effects; its point estimate therefore differs from the table's, and the object of interest is the shape of the set, which is unbounded.

```
ar_high <- iv_ar_confidence_set(high, "y", "gpa", z = "z", x = c("peer", "female"), cluster = "school", weights = "size")
ar_high
#> Anderson-Rubin 95% confidence set (1 instrument(s), n = 20734)
#>   2SLS estimate = 0.2083 (SE 0.2375); Wald interval [-0.2571, 0.6737]
#>   first-stage F (robust) = 2.89; effective F = 2.89
#>   Anderson-Rubin set (unbounded):
#>     [-Inf, 0.6609]
#>     [1.6838, Inf]
```

5 Figure A13: the homogeneity check

The decomposition assumes that the lottery compliers and the students whose grades the peers move share the same γ_2. The paper checks this by re-estimating the second stage in subgroups (Appendix Figure A13). `mediate_iv(homogeneity_by = )` runs the system within each group. In the design γ_2 is homogeneous by construction, so the figure shows what the check looks like when the assumption holds: estimates scattered around one value with intervals that widen with the group's first stage.

```
low$cohort_group <- ifelse(low$cohort <= 2004, "2002-04", ifelse(low$cohort <= 2007, "2005-07", "2008-10"))
low$sex <- ifelse(low$female == 1, "female", "male")
low$program_f <- paste0("program ", low$program)
hom <- bind_rows(lapply(c("sex", "cohort_group", "program_f"), function(g) {
  h <- mediate_iv(low, "y", "peer", "gpa", z = "z", x = "female", fe = c("school", "cohort", "program"),
                  cluster = "school", weights = "size", homogeneity_by = g, n_boot = 0)$homogeneity
  # standard errors of the subgroup mediator effects from the subgroup IV fits
  se <- sapply(h$group, function(lev) {
    sub <- low[low[[g]] == lev, ]
    coeftable(feols(y ~ peer + female | school + cohort + program | gpa ~ z, data = sub, weights = ~size, cluster = ~school))["f
  })
  mutate(h, dimension = g, se = se)
}))
```

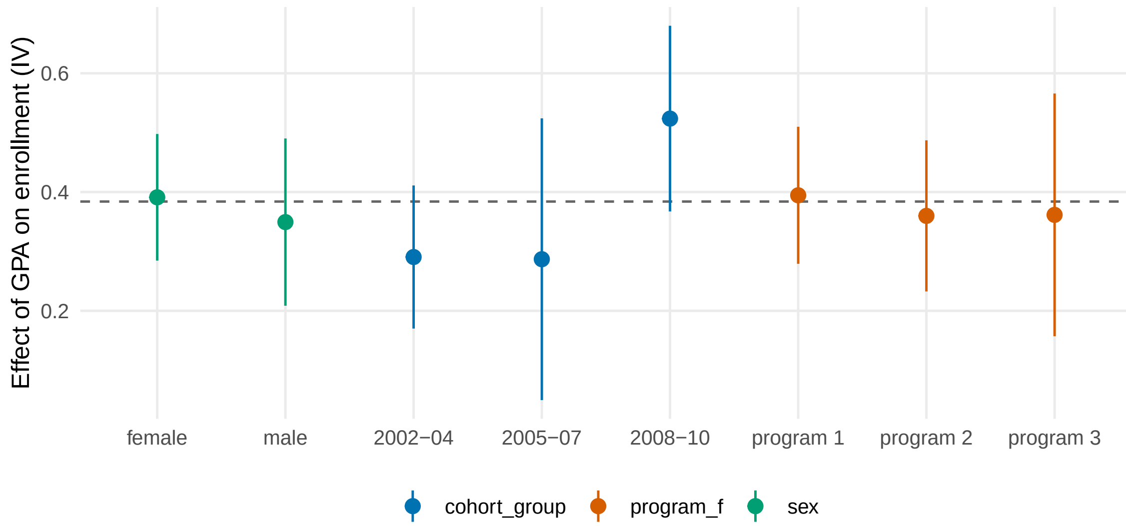

Figure 1: Figure A13 analogue: the instrumented effect of GPA on enrollment by subgroup in the calibrated low-SES design, with 95 percent intervals clustered by school. The dashed line is the design's gamma2 = 0.384.

Table 4: The homogeneity check as a table: subgroup first stages, instrumented GPA effects, and direct effects.

Dimension	Group	N	First stage (peers)	GPA effect	(se)	Direct effect
sex	female	29,019	-0.026	0.391	(0.054)	0.024
sex	male	28,801	-0.035	0.349	(0.072)	0.026
cohort_group	2002-04	19,216	-0.012	0.291	(0.061)	0.029
cohort_group	2005-07	19,171	-0.011	0.287	(0.121)	0.028
cohort_group	2008-10	19,433	-0.057	0.524	(0.080)	0.033
program_f	program 1	19,478	-0.021	0.395	(0.059)	0.027
program_f	program 2	19,225	-0.029	0.360	(0.065)	0.025
program_f	program 3	19,117	-0.043	0.362	(0.104)	0.024

What a failure would look like. If γ_2 differed between, say, programs, the subgroup estimates would separate by more than their intervals, and the product $\delta_1\gamma_2$ would be a weighted average whose weights depend on which students the lottery moves rather than on which students the peers move. The paper's Figure A13 shows estimates that are stable across gender, cohort, and program, and reads that as support for the assumption. It is a necessary check, not a sufficient one: homogeneity across observed groups does not imply homogeneity across the unobserved compliance types.

6 Sampling properties of the decomposition

Repeated draws of the low-SES design show that the system bootstrap is calibrated and that the weak-instrument column is not. For each of 100 draws the direct and indirect effects are estimated with `mediate_iv()` without the bootstrap; the spread across draws is compared with the bootstrap standard error from the first draw.

```
mc <- bind_rows(lapply(seq_len(n_sims), function(s) {
  dl <- sim_cattan(seed = 100 + s)
  dh <- sim_cattan(n_schools = 409, mean_size = 49, delta1 = -0.010, delta2 = 0.043, gamma1 = 0.049, gamma2 = 0.384, seed = 500
  fl <- mediate_iv(dl, "y", "peer", "gpa", z = "z", x = "female", fe = c("school", "cohort", "program"), cluster = "school", wei
  fh <- mediate_iv(dh, "y", "peer", "gpa", z = "z", x = "female", fe = c("school", "cohort", "program"), cluster = "school", wei
  tibble(sim = s, sample = rep(c("low SES", "high SES"), each = 3), term = rep(c("direct", "indirect", "mediator_effect"), 2),
```

```
        estimate = c(fl$estimate[match(c("direct", "indirect", "mediator_effect"), fl$term)], fh$estimate[match(c("direct", "in
}))
```

Table 5: Sampling distribution of the decomposition over 100 draws of each design, against the design values and the school-bootstrap standard errors of the first draw.

Sample	Effect	Truth	Mean across draws	Median	SD across draws	Bootstrap SE (draw 1)
low SES	direct	0.026	0.026	0.026	0.003	0.003
	indirect	-0.015	-0.015	-0.015	0.004	0.004
	mediator_effect	0.384	0.389	0.390	0.043	0.045
high SES	direct	0.049	0.049	0.049	0.008	0.076
	indirect	-0.004	-0.004	-0.003	0.008	0.076
	mediator_effect	0.384	0.401	0.403	0.405	4.606

In the low-SES design the estimator is centred and the bootstrap matches the sampling spread; in the high-SES design neither holds. The mean of the low-SES direct effect across draws is 0.026 with a spread of 0.003, against a bootstrap standard error of 0.003. The high-SES instrumented GPA effect is roughly centred (mean 0.401, median 0.403, design value 0.384) but its spread is 0.405, ten times the low-SES one. The school bootstrap is not a remedy there: the first draw's bootstrap standard errors (0.076 for the direct effect, 4.606 for the GPA effect) are many times even that spread, because replicates in which the first stage lands near zero produce arbitrarily large ratios. A weak first stage has to be reported as such, which is what the paper's F of 5.6 does; its high-SES decomposition (direct 0.049, indirect −0.018) should be read with that column's first stage in mind.

7 What the replication says

Nothing in the paper's Table 4 can be recomputed without the registries, and everything in its design can be. The do-files fix the specification exactly, and the calibrated design reproduces the layout, the magnitudes, and the two features the paper relies on: a strong lottery for low-SES students and a weak one for high-SES students.

`mediate_iv()` is the paper's three regressions plus the standard errors the paper leaves out. The direct effect is the second-stage peer coefficient, the indirect effect is the product, and the total effect is the reduced form; the school bootstrap of the system gives the decomposition's standard errors, and the homogeneity argument re-estimates the system by subgroup.

Two audit findings. The Table 4 IV line carries probability weights without a cluster option, so its reported standard errors are robust rather than school-clustered as the note states; and the "simulated data" one might hope for in `simulate_replication.do` is the Section VII reallocation exercise, which needs the same registry file as everything else.

7.1 Mapping to the lecture's checklist

1. Estimands: total effect of peer exposure, direct effect holding GPA, indirect effect through GPA; identified by the lottery plus homogeneity.
2. Graph: peers to GPA to enrollment, peers to enrollment directly, ability confounding GPA and enrollment, lottery to GPA only.
3. Total effect and the effect on the mediator first: Table 2 and Panel A.
4. Natural effects need an instrumented mediator here: `mediate_iv()`.
5. Sensitivity: to the homogeneity assumption, via subgroups (Figure A13); to the exclusion of the lottery, argued from the blind grading.
6. Instrument: the written-exam lottery; weak in the high-SES sample.

7. No treatment-induced confounder is modelled.
8. No regression accounting.

8 References

- Cattan, Sarah, Kjell G. Salvanes, and Emma Tominey. 2025. "First-Generation Elite: The Role of School Social Networks." *American Economic Review* 115(12), 4369-4403. Replication package: openICPSR.
- Dippel, Christian, Andreas Ferrara, and Stephan Heblich. 2020. "Causal Mediation Analysis in Instrumental-Variables Regressions." *Stata Journal*.
- Huber, Martin. 2019. "A Review of Causal Mediation Analysis for Assessing Direct and Indirect Treatment Effects." IZA Discussion Paper 12660.